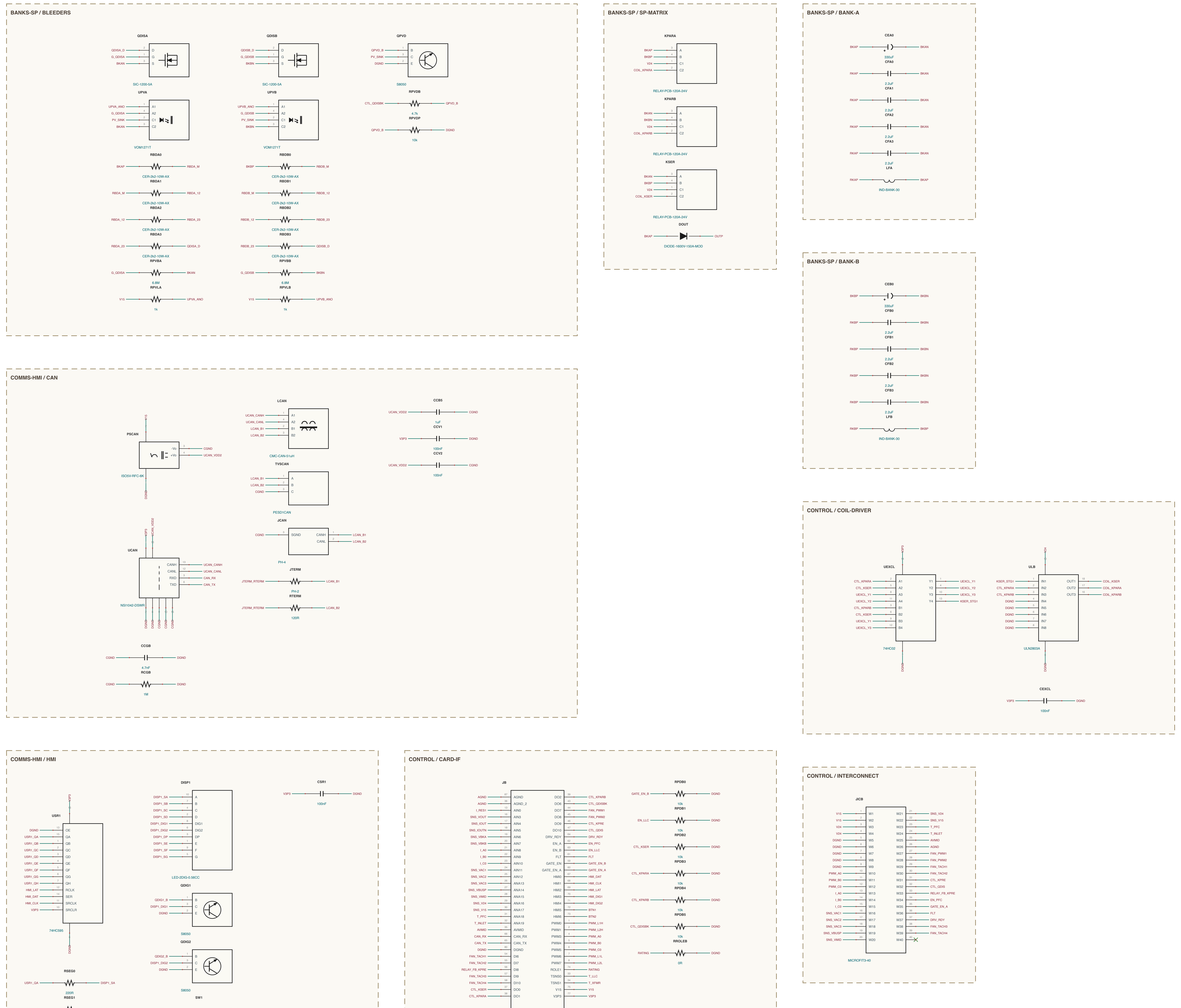


SHEET INDEX

30 kW DC-DC (upper) - 2 of 3

- OUTPUT-SENSING - 6 sections
- BANKS-SP - 4 sections
- LLC-TANKS - 4 sections
- CONTROL - 3 sections
- LLC-LEGS - 3 sections
- COMMS-HMI - 2 sections

rev D.4 - 22 sections - 256 components

NET NAMING

internal junctions are named for what they join

U<ref>_<PIN> node at that IC pin e.g. UIVOA_VINP

R<stem>_M midpoint of a series pair e.g. RBALTA_M

R<stem>_<nm> tap between R<stem>n/m e.g. RV1D_01

all others are explicit design nets

MAGNETICS CONSTRUCTION

identity per docs/magnetics.md (turns, gap, litz, acceptance lines live there)

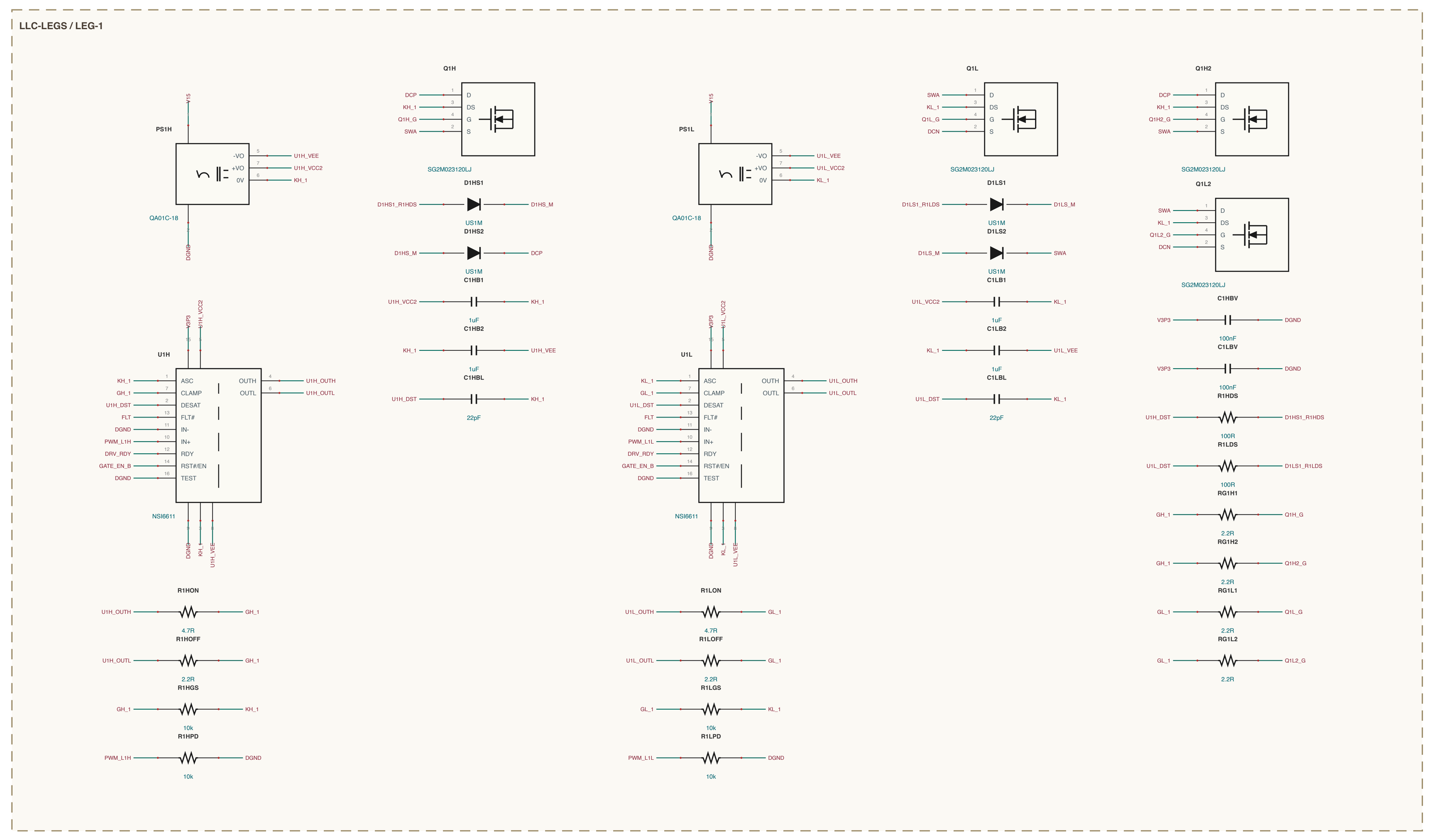
D3 T1A/T1B (E67 rev D): 2 cells, primaries in series -> n 2; each 2x E70/33/32 (B66372B2000), 6:6ll6, pri litz 3850x0.063 + sec foil 0.10x28 halves, Lm 28uH/cell +/-7%, Bpk 159mT, BONDED

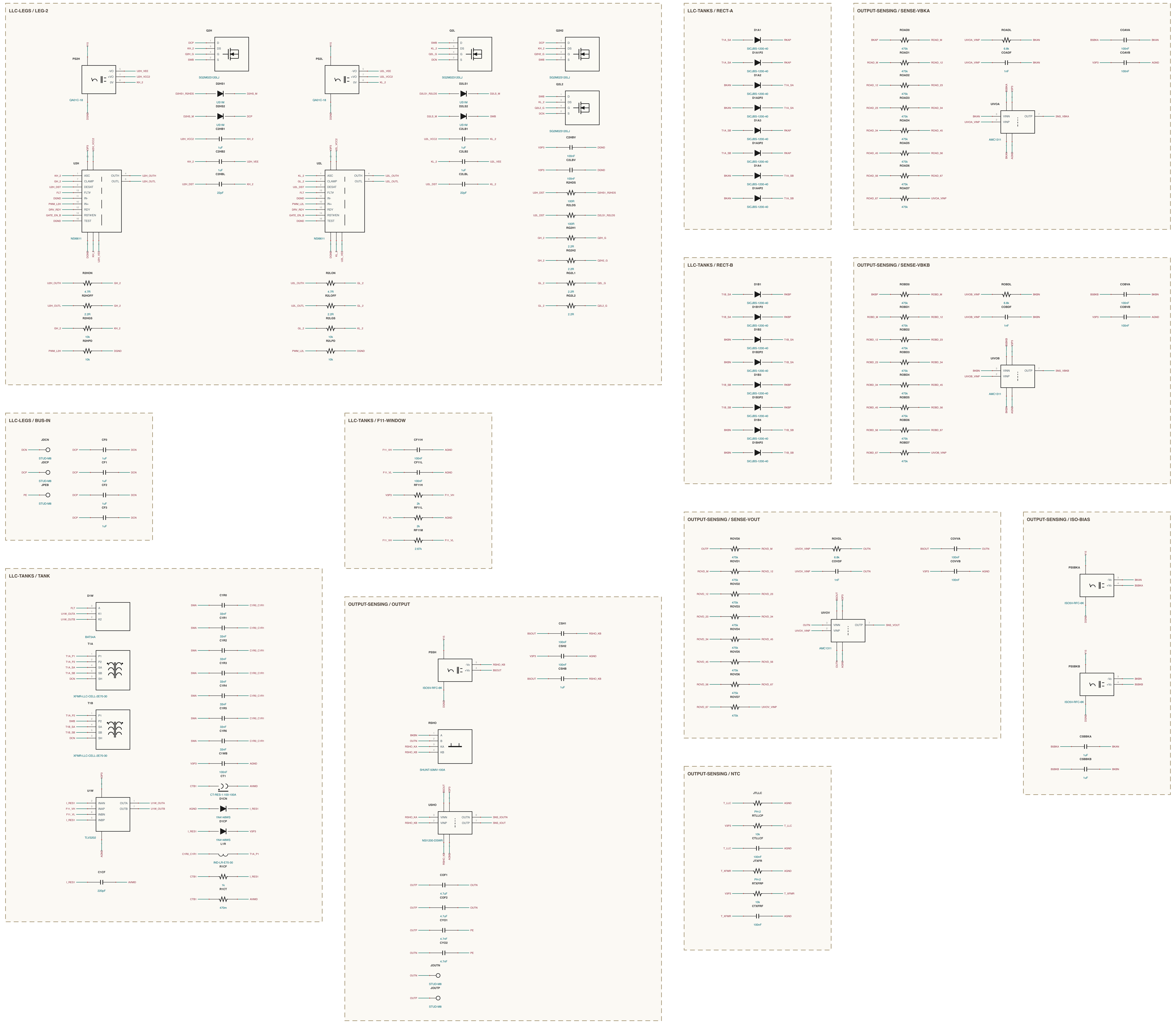
D2 L1R (E67 rev F): external Lr 5.16uH +/-3%, 2x E70/33/32, N=5, litz 8000x0.05, distributed gap, bonded

D8 LFA/LFB: bank filter = D6-30 construction at DC duty, 2x T48 60u N=7 foil 20mm2 -> 7.4uH

total Lr = D2 + 2x cell leakage (acceptance +/-30%) + 0.1uH loop = +/-5% simulated; Cr at TANK

bond loss screened by the EOL bonded thermal soak (T_XFMR NTC rise at fixed load, E65)





DC-Modules 30 kW — DC-DC board (3-phase LLC)

DC-Modules 30 kW - board DC-DC (upper), sheet 2 of 3

2026-09-06 · rev D.4